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Experiment: 8

Steering Angle Variation in MIMO Beamforming Using MATLAB

OBJECTIVE 1:

CODE:

```
clc;
```

```
clear;
```

```
close all;
```

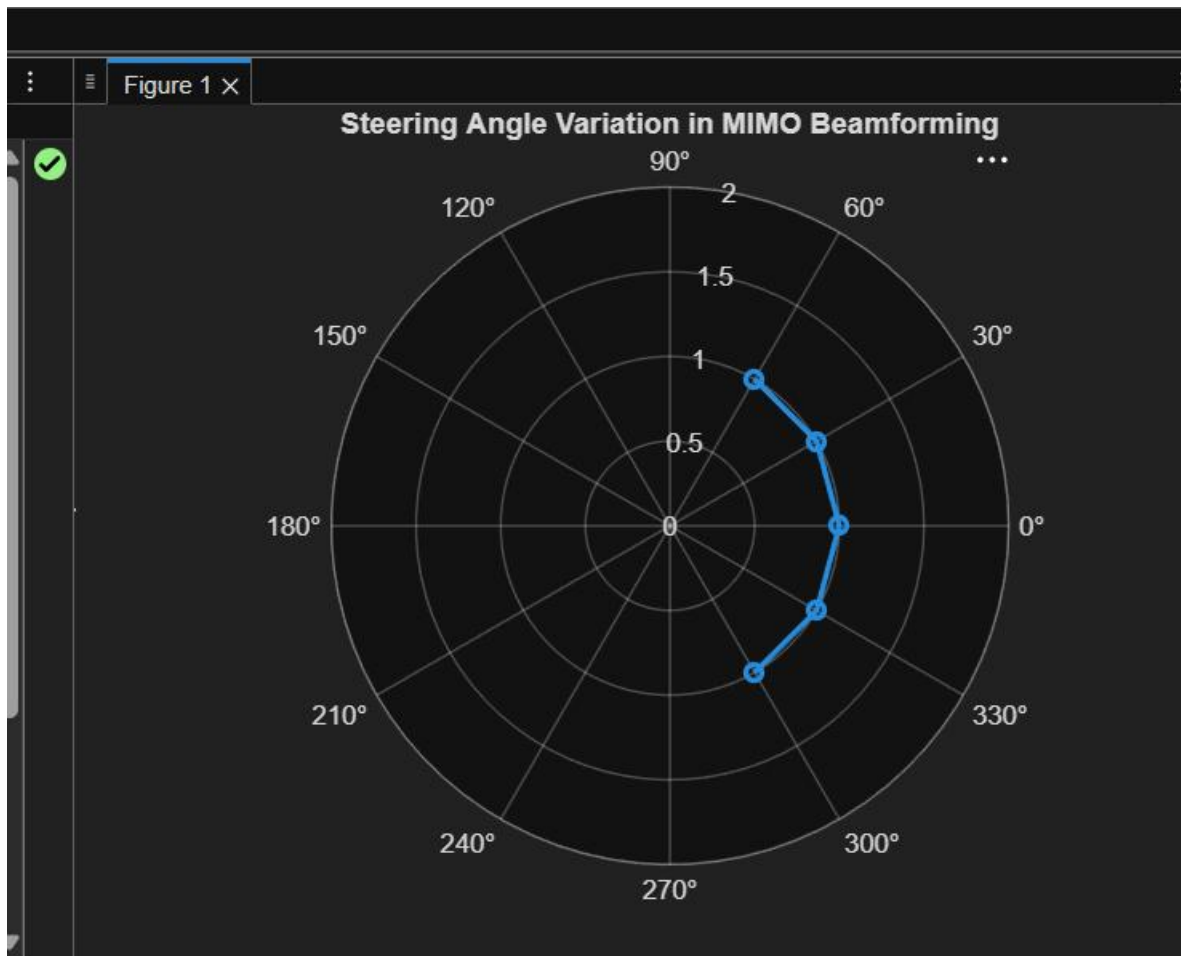
```
angle = [-60 -30 0 30 60]; % Steering Angles (degrees)
```

```
gain = ones(size(angle)); % Normalized Gain
```

```
figure
```

```
polarplot(deg2rad(angle), gain, 'o-', 'LineWidth', 2)
```

```
title('Steering Angle Variation in MIMO Beamforming')
```



OBJECTIVE2:

CODE:

```
clc;
```

```
clear;
```

```
close all;
```

```
angle = [-60 -30 0 30 60]; % Steering Angles (degrees)
```

```
gain = [9 12 15 12 9]; % Array Gain (dB)
```

```
figure
```

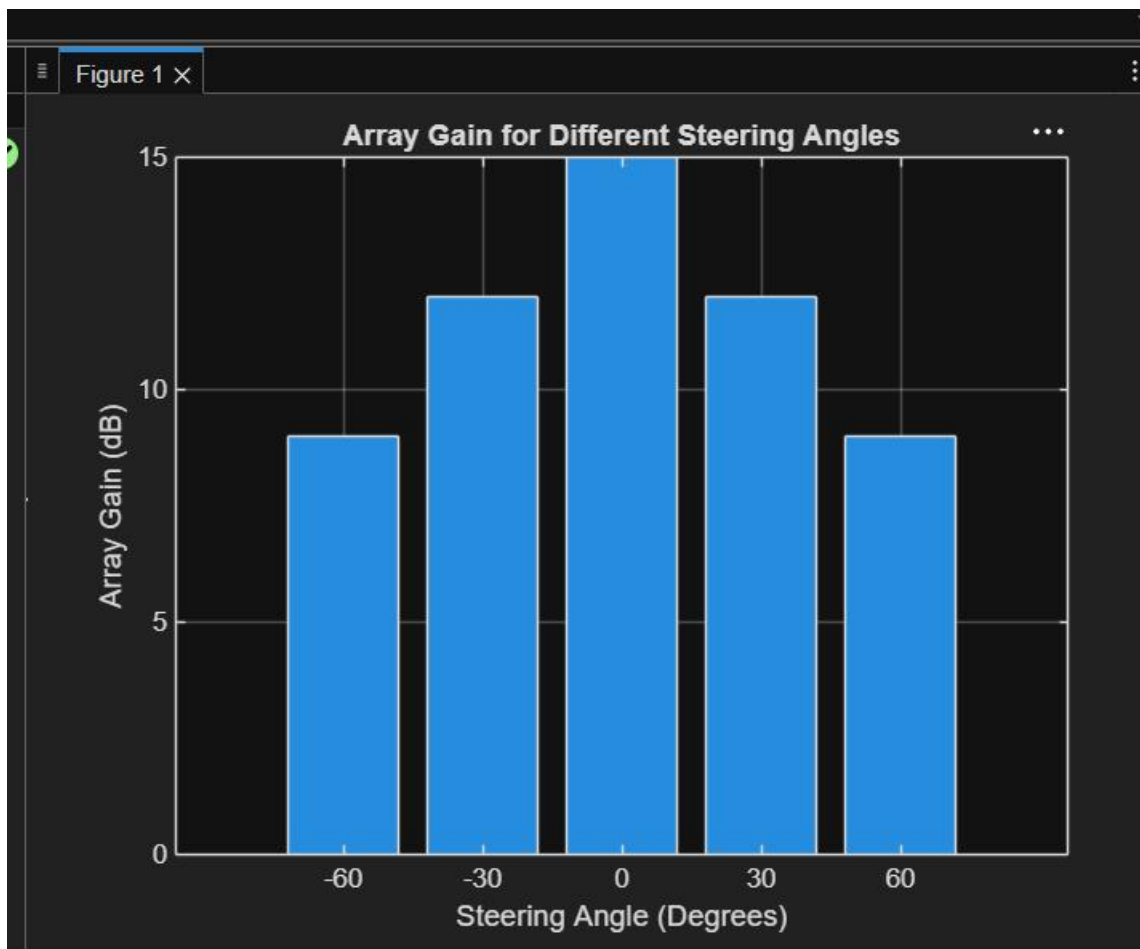
```
bar(angle, gain)
```

```
grid on
```

```
xlabel('Steering Angle (Degrees)')
```

```
ylabel('Array Gain (dB)')
```

```
title('Array Gain for Different Steering Angles')
```



OBJECTIVE 3:

CODE:

```
clc;
```

```
clear;
```

```
close all;
```

```
desired = [-60 -30 0 30 60];    % Desired Steering Angles (degrees)
```

```
actual = [-58 -32 1 29 62];    % Actual Beam Directions (degrees)
```

```
error = abs(desired - actual);  % Beam Direction Error
```

```
figure
```

```
plot(desired, error, 'o-', 'LineWidth', 2)
```

```
grid on
```

```
xlabel('Desired Steering Angle (Degrees)')
```

```
ylabel('Beam Direction Error (Degrees)')
```

```
title('Beam Direction Error for Different Steering Angles')
```

Figure 1 ×

